

Cell Design Specification — V10a_final_h45 (extreme-cold equipment battery)

1. Basic specification

Item	Value	Source
Electrochemical system	NMC811 / graphite (baseline teaching parameterization)	anchor table mapping, entry 0 (task text named no system -> default Chen2020)
Electrolyte	EC/EMC + LiPF6 base (Chen2020); transport design overrides: sigma = 4.0 S/m, D = 3.539e-10 m2/s	params_r6_v10a_h45.json (sigma/D: literature-derived estimates, logged in R3/R4 propose)
Cation transference number	0.45	params_r6_v10a_h45.json
Nominal capacity	5.0 Ah (parameter set) / 5.0618 Ah (1C DFN simulation)	Chen2020 set "Nominal cell capacity [A.h]"; r6_final_1c_dfn.json:capacity_ah
Voltage window	2.5 – 4.2 V	Chen2020 set lower/upper cut-off
Cell dimensions	electrode sheet 65 mm height x 1580 mm width x 184.8 um stack thickness	Chen2020 set height/width; r6_final_energy_dfn.json:thickness_m (wound/pouch form factor not parameterized — "Not provided")

2. Electrode and separator

Layer	Material	Thickness (um)	Porosity	Density (kg/m3)	Source
Positive electrode	NMC811	75.6	0.40 (design override, base 0.335)	3262	Chen2020 set + params_r6_v10a_h45.json
Negative electrode	graphite	85.2	0.35 (design override, base 0.25)	1657	Chen2020 set + params_r6_v10a_h45.json
Separator	polyolefin	10 (base 12)	0.47	397	Chen2020 set + params_r6_v10a_h45.json
Positive current collector	Al	8 (base 16)	—	2700	params_r6_v10a_h45.json
Negative current collector	Cu	6 (base 12)	—	8960	params_r6_v10a_h45.json
Particle radii	pos 3 um / neg 2 um	design overrides			params_r6_v10a_h45.json
N/P ratio	Not provided — Chen2020 exposes no per-electrode capacity-density keys; electrode thickness balance unchanged from baseline (not a design lever this case)				

3. Process design parameters

Parameter	Formula	Value	Unit	Source
Positive areal density	thickness x (1-porosity) x density	147.96	g/m2	75.6e-6 x 0.60 x 3262 (mechanical)
Negative areal density	thickness x (1-porosity) x density	91.76	g/m2	85.2e-6 x 0.65 x 1657 (mechanical)
Positive compaction density	density x (1-porosity) / 1000	1.957	g/cm3	3262 x 0.60 / 1000 (mechanical)
Negative compaction density	density x (1-porosity) / 1000	1.077	g/cm3	1657 x 0.65 / 1000 (mechanical)
Electrolyte fill amount	pore volume x electrolyte density x fill factor	7.98 (fill factor 1.0)	g	pore volume 6.651 cm3 x 1.2 g/cm3 (electrolyte density = literature value, annotated)
Formation recommendation	0.1C CC to 4.2 V, 25 C, 2 cycles	design recommended value; actual production-line value requires tuning		annotated

4. Mass breakdown (contract caliber: layer solids, electrolyte excluded)

Layer	Mass (g)	Source
Positive electrode (solids)	15.196	r6_final_energy_dfn.json:layer_kg_m2.positive_electrode x area 0.1027 m2
Negative electrode (solids)	9.424	layer_kg_m2.negative_electrode x area
Positive current collector (Al)	2.218	layer_kg_m2.positive_cc x area
Negative current collector (Cu)	5.521	layer_kg_m2.negative_cc x area
Separator	0.216	layer_kg_m2.separator x area
Total	32.576	r6_final_energy_dfn.json:mass_kg (electrolyte_included: false)
Electrolyte (estimate, not in contract mass)	7.98	pore volume x 1.2 g/cm3 (literature density, annotated)

5. Performance verification (vs entry-0 criteria)

Metric	Threshold (entry 0)	Value	Verdict	Source
-20 C 1C retention	>= 0.95	0.9944 (5.0337 / 5.0618 Ah)	PASS	derived_r6_final_dfn.json (mechanical lowT/1C)
Energy density	>= 327.18 Wh/kg	564.84 Wh/kg	PASS	r6_final_energy_dfn.json:energy_density_w_h_kg
Volumetric energy density	>= 880 Wh/L	969.50 Wh/L	PASS	r6_final_energy_dfn.json:energy_density_w_h_l
4C/45C max temperature	<= 333.15 K	329.33 K	PASS	r6_final_4C45_safety_dfn.json:T_max_K
4C/45C lithium plating	plated = false	false (anode min +28.96 mV)	PASS	r6_final_4C45_safety_dfn.json:anode_potential_v_min
Robustness re-check (OKane2022 base)	same thresholds	retention 0.9788; T_max 331.78 K; anode min +24.96 mV	PASS	derived_r6_final_ok_spme.json; r6_final_ok_4C45_safety_dfn.json
Cold-start honesty (Initial temp 253.15 K)	retention >= 0.95	0.9944 (T_max 257.10 K confirms cold start)	PASS	derived_r6_final_coldstart.json

6. Design notes (change rationale, citing evaluate log)

1. R1 baseline failed ED_vol (844.45 Wh/L < 880) and 4C plating -> architecture scale (Stage 3).
2. R2/R3: thinned current collectors 16/12 -> 8/6 um and separator 12 -> 10 um (ED_vol), raised anode porosity 0.25 -> 0.30, later 0.35 (plating margin at separator interface) — plating persisted until deep transport was applied.
3. R4: deep transport (sigma 4.0 S/m, t+ 0.45, D 3.539e-10 m2/s, particles 3/2 um, cathode porosity 0.40) + cooling h = 40 W/m2K -> all five criteria pass at SPMe (T_max 330.67 K, anode +22.6 mV).
4. R5: DFN verification passes (T_max 330.47 K, anode +30.1 mV); cold-start honesty check passes; OKane2022 robustness at h=40 fails T_max by 0.165 K (333.315 K).
5. R6: h = 45 W/m2K adopted -> passes both bases at DFN grade (Chen2020 T_max 329.33 K / anode +28.96 mV; OKane2022 T_max 331.78 K / anode +24.96 mV). Final design = V10a_final_h45.